

System and Complexity Science:

Generation, Persistence, and Evolution of Order

— The Physics Constitution for Open Complex Giant Systems

Unabridged Scientific Monograph Edition (v2)

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Abstract

For over two centuries, the governance of open complex giant systems (OCGS) has suffered from the reductionist paradigm collapse under factorial complexity $\mathcal{O}(N!)$ and Wolfram computational irreducibility. In this monograph, we step back to absolute first principles using Occam's razor. We establish a rigorous, minimal axiomatic foundation: starting from Disorder (the unpartitioned whole) and Observer (the self-sustaining steady state), we derive the Triple Realities (Real, Ideal, Residual) and prove the Three Laws of Order: (1) Law of Generation: Without an Ideal Framework (Ω), order collapses ($\Delta S \rightarrow \infty$); (2) Law of Preservation: Without Deterministic Constraints ($\mathbf{\Pi}_\perp$), order dissipates; (3) Law of Evolution: Without Residual Feedback (Δ), evolution stalls into dogma. We formalize these into five formal work operators: Prior Partition Operator $\mathbf{\Pi}$, Rigid Manifold Operator $\mathbf{\Pi}_\perp$, Residual Norm Difference Δ , Second-Order Meta-Cognitive Evolution Operator Φ , and Conscience Tight-Support Operator $\mathbf{E}_{\text{supp}}$. Using Banach Contraction Mapping and Priority Netting, we prove that factorial complexity $\mathcal{O}(N!)$ is algebra-pruned to polynomial solvable $\mathcal{O}(N \log N)$. Finally, we report a 13-year empirical validation across global discrete manufacturing networks (scheduling 2,000,000 parts and 500,000 daily orders in 5 minutes, 98% response rate, 1.9x inventory turnover, human-out-of-the-loop self-healing), establishing a 1:1 mathematical isomorphism across 19 historical philosophies from Laozi and Wang Yangming to Qian Xuesen's Giant Systems, Wu Xuemou's Pansystems, and Longbing Cao's Non-IIDness.

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Writing Constitution: Seven Iron Rules

- **Rule 1: No Overdraft ($\mathcal{O}$ -Overdraft).** Every new term must strictly emerge from previously defined terms.
- **Rule 2: Shortest Path (Geodesic Logic).** From original axioms to realistic inferences, no logical leap is permitted.
- **Rule 3: Minimal Vocabulary (Lexical Sparsity).** Use the simplest, most everyday, self-evident words.
- **Rule 4: Physical Anchoring.** The terminus of the axiomatic chain must connect to real-world engineering data.
- **Rule 5: Homomorphic Preservation.** Explain predecessors using axioms rather than proving oneself using predecessors.
- **Rule 6: Orthogonal Construction.** A complete axiomatic system must first be built using the Three Laws.
- **Rule 7: Falsifiability.** Clear deadlines of falsifiability must be designated across the book.

Preface: Practice Origin and Rectification

The birth of this axiomatic system did not originate from empty speculation, but from a 22-year trajectory of practice, reflection, and scientific elevation. From 2007 to 2020, I served as Chief Architect at Lenovo Group. During those 13 years, my team and I accomplished a historic feat: building an end-to-end, closed-loop autonomous decision-making supply chain planning system—IPS (Integrated Planning Solution). Operating across global manufacturing networks, including the Lighthouse Factory, delivery response rate rose from 54% to 98%, order delivery accuracy improved by +32%, overall inventory turnover increased by 1.9 $\times$, releasing billions of RMB in liquidity.

Introduction: The Origin of Order and Meta-Cognitive Algorithms

Human exploration over millennia is the process of observers partitioning boundaries to construct order. Facing factorial complexity $\mathcal{O}(N!)$ in open complex giant systems, single-point reduction or macroscopic description fails to achieve calculable governance. This monograph constructs a formal closed loop of "Partitioning Generation, Rigid Preservation, Residual Evolution."

Part I

Volume I: Order — First Principles of Open Complex Giant Systems

Chapter 1

Definitions and Axioms

1.1 Observer and Partition

Without an observer, there is no world. Once disorder is observed, it manifests as the world.

Definition 1.1.1 (Disorder). *The unpartitioned whole prior to any observer boundary is defined as Disorder.*

Definition 1.1.2 (Observer). *Within disorder, any entity capable of maintaining a self-sustaining boundary separating internal from external is an Observer.*

Axiom 1.1.1 (Zeroth Law – Observer Emergence Law). *The Observer is a natural self-sustaining steady state emerging within disorder. Without an observer, there is no world.*

Definition 1.1.3 (Partition and Order Construction). *The observer's action of dividing boundaries using Awareness is Partition (Π); establishing rules within the boundary using Construction is Order Construction (Ω).*

1.2 Real, Ideal, and Residual

Partitioning manifests Triple Realities: Real (Ξ), Ideal (Ω), and Residual (Δ).

1.3 The Three Laws of Order

Axiom 1.3.1 (First Law – Generation Law). *Without an Ideal Framework (Ω) established by partition, order collapses ($\Delta S \rightarrow \infty$).*

Axiom 1.3.2 (Second Law – Preservation Law). *Without Deterministic Constraints ($\Pi_{\perp}$), internal elements collide and order dissipates into friction.*

Axiom 1.3.3 (Third Law – Evolution Law). *Without identifying Residual Feedback (Δ), inter-generational evolution stalls into dogma.*

Chapter 2

Order Element Theorem and Dynamic Essence

Theorem 2.0.1 (Order Element Theorem). *Order sustains and evolves if and only if it simultaneously satisfies:*

$$OrderElementTheorem \iff \Pi \text{ (No Collapse)} \wedge \Pi_{\perp} \text{ (No Dissipation)} \wedge \Delta \text{ (No Stagnation)} \quad (2.1)$$

Chapter 3

Formal Outlook and Falsifiability Boundaries

This theory declares three explicit falsifiability deadlines: (1) Order sustains stably without an observer; (2) Collaboration occurs without friction in the absence of constraints; (3) Intergenerational evolution occurs while residual remains identically zero.

Chapter 4

Isomorphism across 19 Historical Philosophers and Scientists

We establish a 1:1 isomorphism across 19 historical paradigms from Laozi, Wang Yangming, Kant, to Qian Xuesen, Wu Xuemou, and Longbing Cao.

Part II

Volume II: Formal Work Operators and Algorithms

Chapter 5

Ideal World Evolution and the 400-Year Crisis of Reductionism

Reductionism decomposed complex systems into independent micro-units, collapsing under factorial complexity $\mathcal{O}(N!)$.

Chapter 6

Paradigm Shift and Calculable Work

Algebraic pruning via operator Π and residual-driven paradigm shifts via operator Φ achieve calculable closed-loop governance.

Chapter 7

Formal Work Operators and Topological Manifolds

Prior Partition Operator $\mathbf{\Pi} = \langle \mathcal{D}, A \rangle : \Xi \rightarrow \Omega$, Rigid Manifold Operator $\mathbf{\Pi}_\perp$, and Residual Norm Difference $\Delta = \|\Omega_t - \Omega_{t-1}\|$.

Chapter 8

Embodied Operators and Neural Topologies

Theorem 8.0.1 (Banach Contraction and Decidability). *Let $T = \Pi_{\perp} \circ \Phi$. Then with $d(Tx, Ty) \leq \gamma d(x, y)$, $\gamma < 1$, guaranteeing convergence to a Pareto-optimal state and reducing factorial complexity $\mathcal{O}(N!)$ to polynomial $\mathcal{O}(N \log N)$ via priority netting.*

Part III

Volume III: Physical Constitution and Dimensionality Reduction Codex

Chapter 9

Teleology: Three Physical Pillars of Holographic Anti-Entropy

Vector cancellation equation:

$$V_{\text{sys}} = \sum_{i=1}^N V_i, \quad Q_{\text{waste}} = \sum_{i=1}^N \|V_i\| - \left\| \sum_{i=1}^N V_i \right\| \quad (9.1)$$

Eliminating non-orthogonal components forces vector cancellation waste heat to zero ($Q_{\text{waste}} \rightarrow 0$).

Chapter 10

Ontology: Human-Machine Collaboration

Node interactions explode factorially Complexity $\sim \mathcal{O}(N!)$.

Chapter 11

Scheme Theory: Five-Dimensional Double Helix

Nodes (N), Topology ($\mathbf{A}$), Constraint Cluster (C_{st}), State Transitions (S), State Vectors ($\mathbf{x}(t)$).

Chapter 12

Capability Theory: Rigid Manifold Stripping

Stripping normal redundant degrees of freedom: $x_{\perp}(t) = x(t) - \Pi_{\perp}x(t) \rightarrow 0$.

Chapter 13

Mechanism Theory: Decision Write-Back

Control domain dimension $\dim(C) \leq \dim(O)$. Automated decision write-back executes millisecond self-healing.

Chapter 14

Path Theory: Micro-Perception to Macro-Control

Filtering residual $\Delta(t) = \mathbf{\Pi} \cdot (\mathbf{x}_{\text{real}}(t) - \mathbf{x}_{\text{model}}(t))$.

Chapter 15

Work Theory: Info-Energy Conservation

Landauer's Principle $\Delta E \geq k_B T \ln 2 \cdot \Delta I$.

Chapter 16

Evolutionary Theory: Residual Accumulation

$$\Phi : \Pi_k \rightarrow \Pi_{k+1}.$$

Part IV

Volume IV: Cross-Domain Empirical Proof and Symbiotic Civilization

Chapter 17

Cross-Domain Dimensionality Reduction and Information Entropy

Empirical validation across global discrete manufacturing networks proves that physical production entropy is bounded by Prior Partition Operator Π , collapsing $\mathcal{O}(N!)$ down to $\mathcal{O}(N \log N)$ in 296 seconds.

Chapter 18

Silicon-Based Symbiotic Self-Healing Evolution

Operational metrics: 98% response rate, +32% delivery accuracy, 1.9× inventory turnover boost.

Epilogue: The Perpetual Anti-Entropy Engine

Bridging Eastern wisdom and Western rigor under the Holographic Anti-Entropy Constitution.

Appendix A

Core Axiom Derivations and Quotations

A.1 Vector Cancellation Equation and Waste Heat Theorem

$$Q_{\text{waste}} = \sum_{i=1}^N \|V_i\| - \left\| \sum_{i=1}^N V_i \right\| \implies \Delta S \rightarrow \infty.$$

A.2 Goodhart Inequality and Tight-Support Operator Clipping Proof

$$\mathbf{E}_{\text{supp}} \cdot p(\Delta) \in \text{Sub-Gaussian}.$$

A.3 $\mathcal{O}(N!)$ Factorial Complexity Emergence and Pruning Convergence Theorem

Prior partition operator $\Pi = \langle \mathcal{D}, A \rangle$ compresses $\mathcal{O}(N!)$ to $\mathcal{O}(N \log N)$.

A.4 Landauer's Principle and Dissipation Entropy Lower Bound

$$|\Delta S_{\text{sys}}| > S_{\text{ctrl}}.$$

Appendix B

Five-Dimensional Numerical Reconstruction Algorithms

B.1 Residual-Driven Write-Back Instruction Set

Modifying Nodes (N), Topology (T), Constraints (C), State Vectors (V), State Transitions (S).

B.2 Subject Exclusivity of Second-Order Metacognitive Operator and Human Metacognition Theorem

$$\mathcal{O}_{\text{system}} = \Phi_{\text{Human}} \otimes A_{\text{Silicon}}.$$

Appendix C

Topological Isomorphism Map with Contemporary Frontiers

Isomorphism with Prof. Longbing Cao's Non-IID Learning Theory.

Appendix D

Full Logic Flowchart and Phase Space Lookup Table

$$\Pi \rightarrow \Pi_{\perp} \rightarrow \Delta \rightarrow \Phi \rightarrow \mathbf{u}^*(t).$$